Data Analysis Assignment

Problem Statement — Exploratory Data Analysis (EDA) with Python

Duration	2 hours	**Dataset file**	Loan_Dataset.csv
Format	Individual, open-book (Python only)	**Rows / Columns**	772 rows x 17 columns
Tools allowed	Jupyter Notebook / Google Colab / VS Code	**Submission**	1 .ipynb notebook + exported .html
Libraries	pandas, numpy, matplotlib, seaborn	**Language**	Python 3

Scenario

You have just joined the Risk Analytics team of a lending company as a Data Analyst. The Loans team has exported the full loan book directly from the core lending system, covering loans disbursed between 2021 and late 2024. The export has never been cleaned or analyzed. Defaults have been creeping up and the Head of Risk wants a clear, code-driven exploratory analysis before deciding whether to tighten underwriting criteria. You need to show you can take a messy, real-world export and turn it into a structured, well-documented EDA notebook that a non-technical stakeholder could still follow.

The Dataset

You are given **Loan_Dataset.csv**, a single flat file with the following columns:

Column	Description
Loan ID / Customer ID / Customer Name	Identifiers
Gender / Age	Demographic fields
Employment Type	Salaried / Self-Employed / Business Owner
Monthly Income (INR) / Credit Score	Financial profile at time of application
Region / Branch City	Branch location
Loan Type	Personal / Home / Auto / Education / Gold / Business Loan
Loan Amount (INR) / Interest Rate (%) / Tenure (Months)	Loan terms
Disbursement Date	Date the loan was disbursed (formats are inconsistent, see Task 1)
EMI (INR)	Monthly installment amount
Loan Status	Active / Closed / Defaulted / Written Off, as of 31-Dec-2024

Note: the data was deliberately exported without any cleanup. It includes missing values, duplicate rows, inconsistent text casing and spacing, mixed date formats, a handful of invalid (negative) ages, a few extreme EMI outliers, and a few logically inconsistent rows (a loan marked 'Closed' with a disbursement date only weeks old) — all issues you'd encounter in a real core-banking export.

Tasks

Do all work in a single Jupyter Notebook. Structure it with clear markdown section headers matching the task numbers below, so a reviewer can navigate it top to bottom. Code should run end-to-end with Restart & Run All — no cells that only work out of order.

Task 1: Data Loading & Inspection [10 pts]

- Load the CSV with pandas and inspect shape, dtypes, and a sample of rows.
- Produce a summary of missing values per column and the count of exact duplicate rows.
- Report basic descriptive statistics (.describe()) for all numeric columns.

Task 2: Data Cleaning [20 pts]

- Parse Disbursement Date into a proper datetime column despite the mixed formats in the raw file.
- Standardize inconsistent text casing/spacing in Region, Branch City, and Customer Name.
- Remove exact duplicate rows (show the row count before and after).
- Handle invalid Age values (negative numbers) — flag or correct them, with your reasoning stated in a markdown cell.
- Handle missing values in Employment Type, Monthly Income, Credit Score, Branch City, and Interest Rate with a clearly justified method (impute, flag, or drop) — do not silently drop rows without stating why.
- Identify and flag the logically inconsistent rows where Loan Status = 'Closed' but the Disbursement Date is less than 3 months before 31-Dec-2024.
- Add a derived column 'Loan Age (Months)' = months elapsed between Disbursement Date and 31-Dec-2024.

Task 3: Univariate & Bivariate Analysis [20 pts]

- Explore the distribution of Loan Amount, Interest Rate, Credit Score, and EMI (histograms or KDE plots).
- Compare Loan Status across Loan Type, Employment Type, and Region using cross-tabulations (pd.crosstab) and grouped bar charts.
- Compute and visualize the correlation between numeric variables (Credit Score, Monthly Income, Loan Amount, Interest Rate, EMI) using a correlation heatmap.
- Identify outliers in Loan Amount and EMI using a statistical method (IQR or Z-score) and report how many rows are affected.

Task 4: Risk-Focused Analysis [20 pts]

- Calculate the overall default rate (% of loans that are 'Defaulted' or 'Written Off').
- Break down default rate by Loan Type, Employment Type, and Credit Score band (e.g. <650, 650-750, 750+) — bucket Credit Score using pd.cut.
- Analyze whether EMI-to-Income ratio (EMI / Monthly Income) relates to default risk — visualize with a boxplot comparing Active/Closed vs Defaulted/Written Off loans.
- Build a simple rule-based 'high risk' flag (e.g. low credit score AND high EMI-to-income ratio) and report how many currently Active loans fall into it.

Task 5: Visualization Summary [15 pts]

- Produce at least 5 well-labeled charts total across the notebook (title, axis labels, legend where relevant) — reuse from earlier tasks is fine, but they must be polished, not default matplotlib styling.
- Include at least one multi-panel figure (e.g. plt.subplots) summarizing portfolio composition by Region and Loan Type.

Task 6: Business Insights & Recommendation [15 pts]

- In a final markdown section, write 4-6 bullet points of insights a Head of Risk would act on (not just restating numbers).
- Recommend at least one concrete underwriting change (e.g. a credit score cutoff, an EMI-to-income cap, a loan type to review) backed by a specific number from your analysis.

Bonus (optional, up to 5 bonus pts)

Fit a simple logistic regression or decision tree (scikit-learn) predicting default (Defaulted/Written Off vs not) from Credit Score, EMI-to-Income ratio, and Loan Type, and report which features matter most. This is a bonus, not a requirement — a correct, well-reasoned EDA without modeling still earns full marks on Tasks 1-6.

Evaluation Rubric (100 pts total)

Area	Weight	What we're looking for
Data Loading & Inspection	10	Correct, efficient use of pandas for a first look at the data
Data Cleaning	20	Correctness of fixes, sensible and clearly justified handling of missing/bad/inconsistent data
Univariate & Bivariate Analysis	20	Right statistical techniques for the question, correct interpretation, appropriate use of groupby/crosstab
Risk-Focused Analysis	20	Correct default-rate calculations, sensible bucketing, a defensible risk flag
Visualization	15	Charts are correctly chosen, clearly labeled, and easy to read — not default/unstyled output
Insights & Recommendation	15	Specific, decision-relevant insights tied to numbers actually produced in the notebook

Rules

- Do not use any AI tool, browser extension, or another person's help during the test. This round evaluates your own analysis skills.
- You may refer to official pandas/matplotlib/seaborn documentation.
- Every chart and statistic must come from code in the notebook — no numbers typed in by hand.
- Comment your code and use markdown cells to explain your reasoning, especially in Task 2 (Data Cleaning) and Task 6 (Insights).
- The notebook must run top-to-bottom without errors (Kernel: Restart & Run All) before you submit.
- Partial credit is given — attempt every task even if you can't finish it fully.

Submission

Export your notebook with all outputs visible and submit two files: **FirstName_LastName_Submission.ipynb** and a rendered **FirstName_LastName_Submission.html** (File > Download as > HTML, or File > Export) as a backup in case of environment issues. Upload both through the link shared by the recruitment team before the timer ends. Late submissions after the 2-hour window will not be evaluated.